

Political Alpha.

A test of whether U.S. Senate stock disclosures predict sector ETF behavior. And why, mostly, they don't.

The question.

Do U.S. Senate stock disclosures carry predictive signal for forward sector ETF returns, and does the 45-day disclosure delay erode that edge?

WHY IT MATTERS

The STOCK Act (2012) requires senators to disclose trades within 45 days. This project assesses whether that lag removes any predictive signal once trades are public.

APPROACH

Scraped 2,284 Senate reports, tagged trades to sectors, built weekly signals, and regressed forward ETF returns on lagged signals.

PRE-REGISTERED EXPECTATION

With a 6-week disclosure lag the null is no predictive signal; rejection suggests leakage or skill.

Four sources.

Weekly panel, 2023–2026. One of the four (Congress.gov) turned out to be unusable (see §06).

SOURCE	ACCESS	COVERAGE	ROLE
Yahoo Finance	yfinance library	1,507 days × 7 tickers	ETFs, VIX, SPY, ^IRX
Senate EFD	HTTP + CSRF session	2,284 trades · 34 senators · \$207M notional	Trade signal
Congress.gov	REST API with key	5,000 bills pulled, 276 tagged (5.5%)	<i>Legislative signal (unused)</i>
Ken French Library	public ZIP	1,509 daily rows: Mkt-RF / SMB / HML / RF	Factor controls

Five sectors tracked: Defense (ITA) · Finance (XLF) · Energy (XLE) · Healthcare (XLV) · Tech (XLK)

How the pipeline runs.

Four stages. Every step reproducible from one command: `python main.py`

01 **FETCH** Pull raw data

yfinance daily closes; Senate EFD scraper with CSRF/cookies and paginated POSTs; Congress.gov REST and Ken French ZIPs.

02 **PARSE & TAG** Structure + label

Normalize PTR rows (senator, ticker, direction, amount). Sector tagging via small ticker whitelist & keyword dictionary; ~18% of trades tagged.

03 **BUILD SIGNAL** Weekly heat score

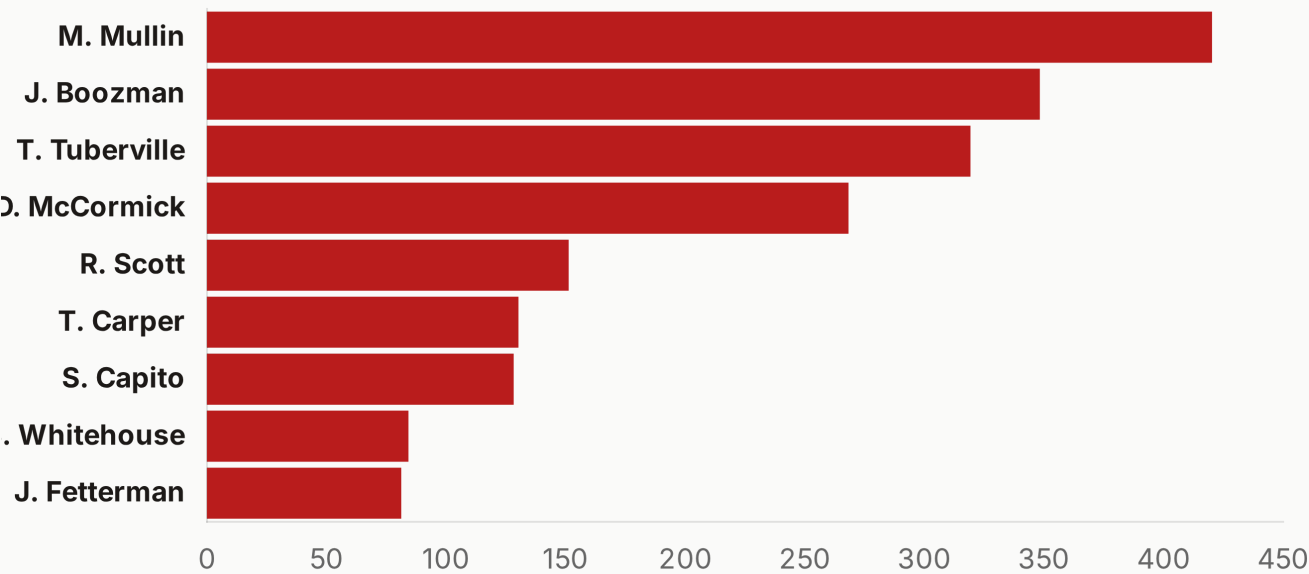
Aggregate to weekly counts; rolling z-score (26w) and 1-week lag to avoid look-ahead.

04 **TEST** Statistical inference

OLS of forward returns on lagged heat (1,5,20d); Newey-West SEs. 30 specifications.

Three senators drove half the activity.

Not a market. Three portfolios.



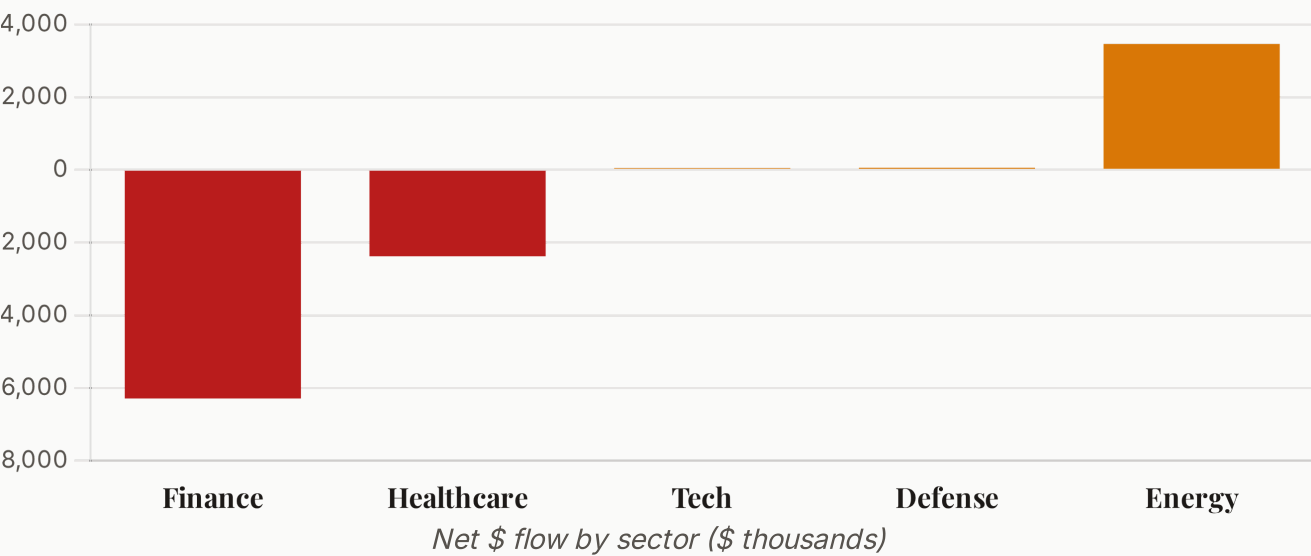
47.6%
of all trades, three senators.

2,284
disclosed Senate trades, 2023–2026.

\$207M
total notional (bucket midpoints).

Senators net-sold healthcare and finance.

Dollar-weighted net flow (buys minus sells) across 419 sector-tagged trades.

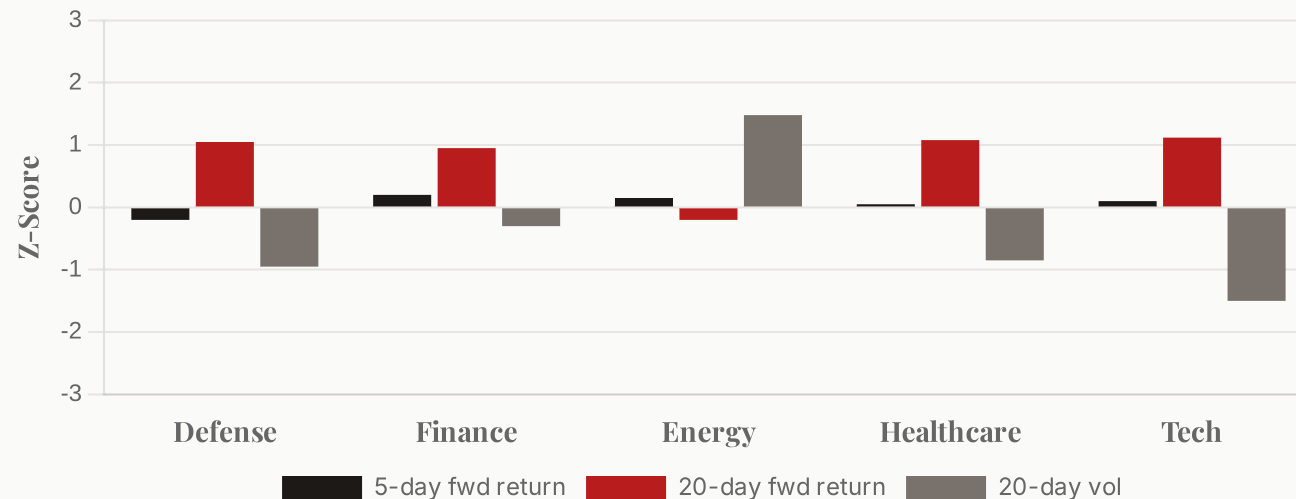


BUY SHARE BY SECTOR

Healthcare	32%	24 buy / 50 sell
Tech	44%	78 / 98
Finance	51%	39 / 38
Energy	53%	43 / 38
Defense	56%	5 / 4

No persistent predictive signal.

OLS · HAC standard errors · sector $\times$ forward window · 1-week lag to prevent look-ahead



HAC t-statistic by sector. The $|t|=1.96$ significance threshold sits just above 2 on the y-axis.

0 / 20

TESTS SIGNIFICANT AT $P < 0.05$

"A secondary specification (raw net-buy count, no z -score) produced one hit, healthcare 20-day, $p = 0.03$. It did not replicate in an out-of-sample split (2nd half $p = 0.46$). Driven by one or two large sales."

The edge has faded.

Our null is consistent with the post-STOCK-Act literature. The 2024 headlines are driven by outliers and holding effects.

1993–2008

Pre-STOCK Act

+12%

per year

Senate portfolios historically beat the market; disclosures were limited.

2012–2020

Post-STOCK Act

~0

abnormal return

Studies find little evidence of persistent stock-picking skill after disclosure.

2024–2025

Recent Headlines

32.2%

of members beat SPY in 2025

Many winners in 2024 held concentrated ETFs (e.g., NVDA exposure) rather than active trading.

Takeaway: *post-disclosure, the active-trading edge largely disappears. Our 2023–2026 null fits the pattern.*

Long-short portfolio generates no significant risk-adjusted return.

Weekly long-short: long top-2 sectors by heat score, short bottom-2. Equal-weighted. n=152 weeks.

CAPM ALPHA		FF3 ALPHA	
alpha (weekly)	-0.000594	alpha (weekly)	-0.000210
alpha (annual)	-3.04%	alpha (annual)	-1.09%
alpha t-stat	-0.438	alpha t-stat	-0.153
alpha p-value	0.661	alpha p-value	0.879
market beta	0.053	market beta	-0.064
R ²	0.003	SMB beta	0.161
		HML beta	-0.416
		R ²	0.137

+4.30% **-15.46%**
Cumulative return Max drawdown

*Neither CAPM nor FF3 alpha is significant ($p \gg 0.05$).
The 4.3% cumulative return is explained by factor exposures, not the congressional trading signal.*

The signal is consistently zero.

We checked if the link between Senate trades and stock prices changed at any point between 2022 and 2026.

THE "BREAK" TEST

Searching for sudden shifts

We looked for any specific week where the trading signal suddenly became more (or less) accurate. This helps ensure we didn't miss a short window where the strategy actually worked.

THE "DRIFT" TEST

Watching for gradual changes

We monitored the data for any slow decay or improvement in the signal over time. This catches subtle trends that a single "break" might miss.

THE RESULT

The data shows no significant "jumps" or "drifts." The predictive power of these trades didn't just fail recently—it has been absent for the entire four-year period we studied. There is no evidence of a "hidden" window of success.

What made this hard.

I Senate EFD scraping

CSRF and cookie handshake required; terms-of-use POST before requests. Many older PTRs are scanned PDFs and unparseable.

II Congress.gov coverage mismatch

API defaults returned largely old housekeeping bills; only a small fraction were sector-tagged and few matched the 2023–2026 window.

III Small, concentrated sample

34 senators report with 3 accounting for 48% of activity. Defense trades are very few, so power to detect effects is low.

IV Multiple-comparison discipline

30 specifications across tests give ~1.5 expected Type-I errors; observed 1. Did not treat the healthcare hit as definitive without out-of-sample checks.

Scaling the search for Alpha.

From aggregate sector flows to high-resolution individual analysis and historical context.

I. DATA EXTRACTION	II. GRANULAR ANALYSIS	III. HISTORICAL CONTEXT
Groq LLM PDF Parsing	Individual Member Focus	The Tipping Point
Use Groq’s LLM parser to extract rows from legacy scanned filings, reducing the 20–40% empty-rate and unlocking older PTR data.	Shift from sector heat scores to member-level performance to identify super-traders and correlate trades with committee activity.	Long-horizon study comparing pre- and post-STOCK Act periods to map when and how the congressional edge decayed.

Goal: *Determine whether the edge vanished or migrated to concentrated, individual strategies.*

On process.

Expectations guided my choices; I then tested them against out-of-sample checks.

WHERE THE BIAS SHOWED UP

Following promising results too early.

I sometimes built a narrative around a single specification before confirming robustness.

WHAT THE PROCESS ADDED

Writing expectations down before testing.

Declaring the null and using out-of-sample checks helped interpret results on their own terms.

Takeaway: *methodology matters more than any single finding; the design made the null informative.*